

## TTRS Computational Results

The displayed formulation below is included only so that this file compiles independently. Copy only the block marked “BEGIN PAPER INSERT” into paper/Lorentz.tex.

$$\min\{x^\top Qx + 2g^\top x : \|x\| \leq 1, \|\text{diag}(a)x + b\| \leq 1\}. \quad (1)$$

We ran Lazy SEP on the 212 centered diagonal TTRS instances from the ball-constraints repository, and Full SEP on 212 instances: all instances with  $n \in \{5, 10, 20\}$ . Shor and KRON are omitted from this panel. The implementation uses the transpose of the paper’s bilinear convention. It forms the square SEP lift

$$Z = \begin{pmatrix} 1 & y^\top \\ x & V \end{pmatrix}, \quad y = \text{diag}(a)x + b,$$

and represents  $V_{ij} = x_i y_j = x_i(a_j x_j + b_j)$  linearly through the  $x$ -moment matrix. Full SEP imposes all skew-skew equations at once, while Lazy SEP adds violated coordinate skew-skew equations by separation. For every SDP row, bound quality is measured using that method’s own recovered first-column feasible point as an upper bound; neither Gurobi nor the corrected beta values are used to calculate an SDP gap.

Table 1: Results on TTRS instances from [?]

| $n$ | cases | Full SEP  |       |                     | Lazy SEP |       |                     | Gurobi   |       |                     |
|-----|-------|-----------|-------|---------------------|----------|-------|---------------------|----------|-------|---------------------|
|     |       | time      | tight | max gap             | time     | tight | max gap             | time     | tight | max gap             |
| 5   | 38    | 13 ms     | 38    | $4.9 \cdot 10^{-7}$ | 25 ms    | 38    | $1.2 \cdot 10^{-6}$ | 49 ms    | 38    | $8.7 \cdot 10^{-7}$ |
| 10  | 70    | 457 ms    | 70    | $2.3 \cdot 10^{-7}$ | 144 ms   | 70    | $2.0 \cdot 10^{-7}$ | 447 ms   | 69    | $1.5 \cdot 10^{-4}$ |
| 20  | 104   | 1016.15 s | 104   | $3.5 \cdot 10^{-7}$ | 2.18 s   | 104   | $7.5 \cdot 10^{-7}$ | 487.45 s | 93    | $1.5 \cdot 10^{-4}$ |

Table 2: Additional diagnostics for Lazy SEP on TTRS instances from [?]

| $n$ | rounds | cuts    | max skew            |
|-----|--------|---------|---------------------|
| 5   | 2/2    | 80/100  | $7.8 \cdot 10^{-9}$ |
| 10  | 2/2    | 100/100 | $8.4 \cdot 10^{-9}$ |
| 20  | 2/3    | 100/200 | $3.9 \cdot 10^{-8}$ |